

SQL Research Project



Student Performance Analysis Using SQL & Excel

An Analytical Study of the Factors Associated with Academic Performance



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Introduction

Student academic performance can be influenced by various academic, personal and environmental factors. This project analyzes student performance data using SQL and Excel to identify patterns, compare groups and understand factors associated with Final Percentage.

Student Data

SQL Analysis

Excel Visualization

Research Insights



Research Objectives

To analyze student academic performance and identify the key factors influencing students' overall performance and success.



Research Problems Addressed

Study Hours

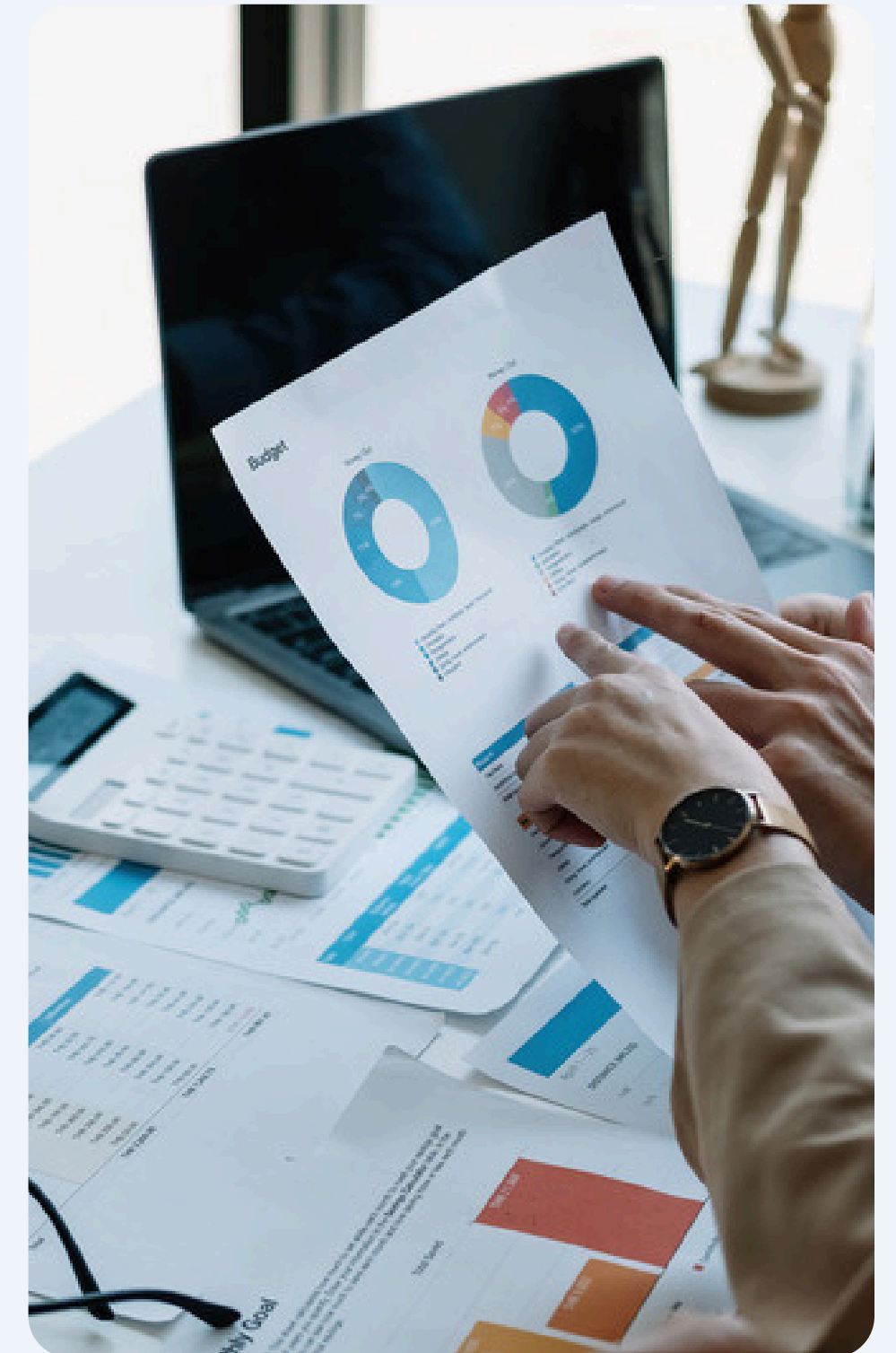
Attendance

Internet Access

Parental Education Access

Class-wise Ranking

Overall Performance





Dataset Overview

Number of Students : 5000

Number of Columns : 16

Primary Key: Student_ID

Category	Variables
Demographics	Age, Gender, Class
Study Habits	Study Hours, Attendance
Environment	Parental Education, Internet Access
Academic	Math, Science, English
Performance	Previous Score, Final %, Performance Level, Pass/Fail

Research Methodology



Dataset



Data Preparation



MySQL Database



Research Findings



SQL Queries



Statistical Analysis



Excel Visualization

Overall Student Performance

What is the average Final Percentage of all students?

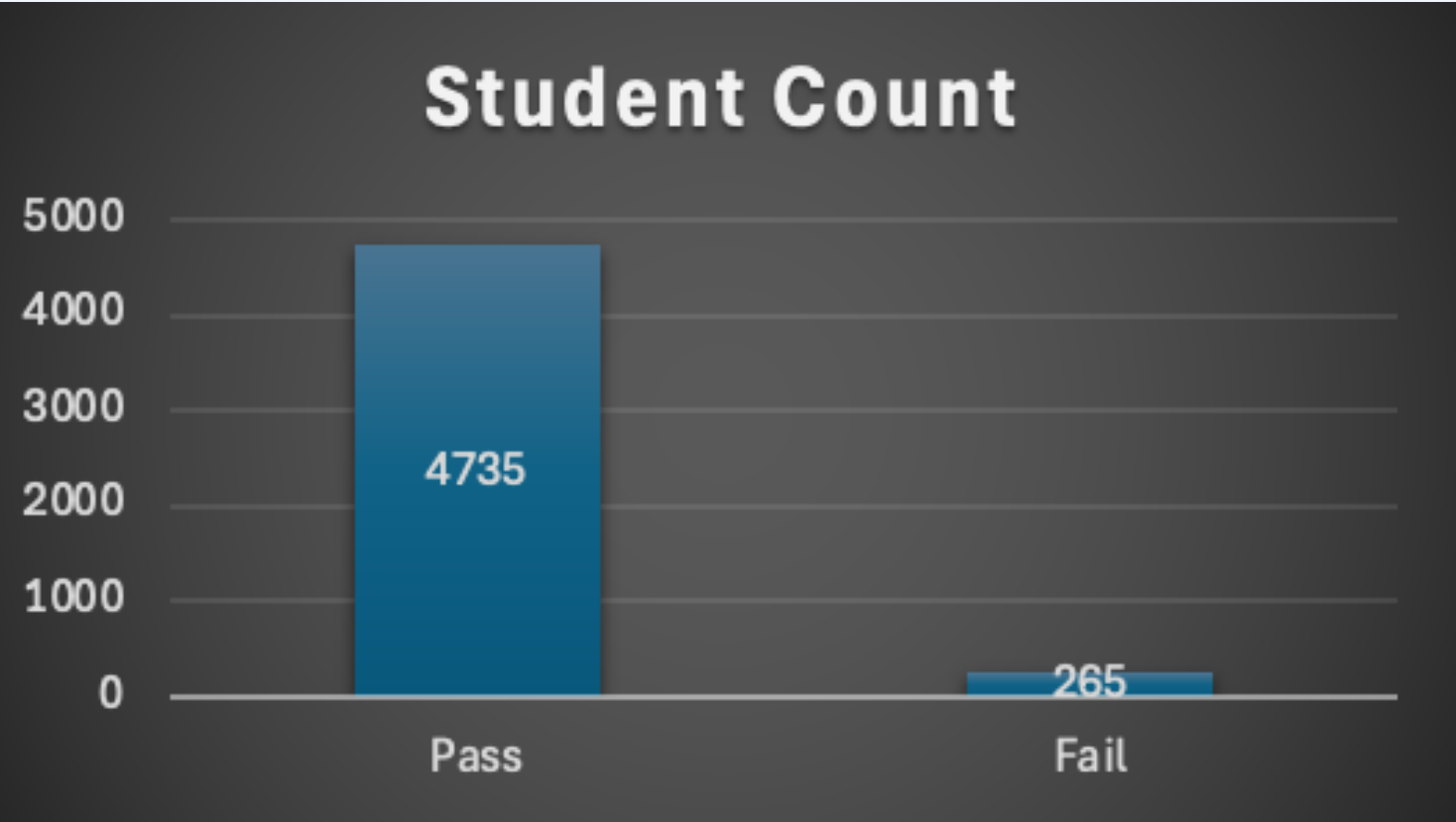
Average Final Percentage

67.48%

The overall average Final Percentage provides a baseline for evaluating individual and group performance.

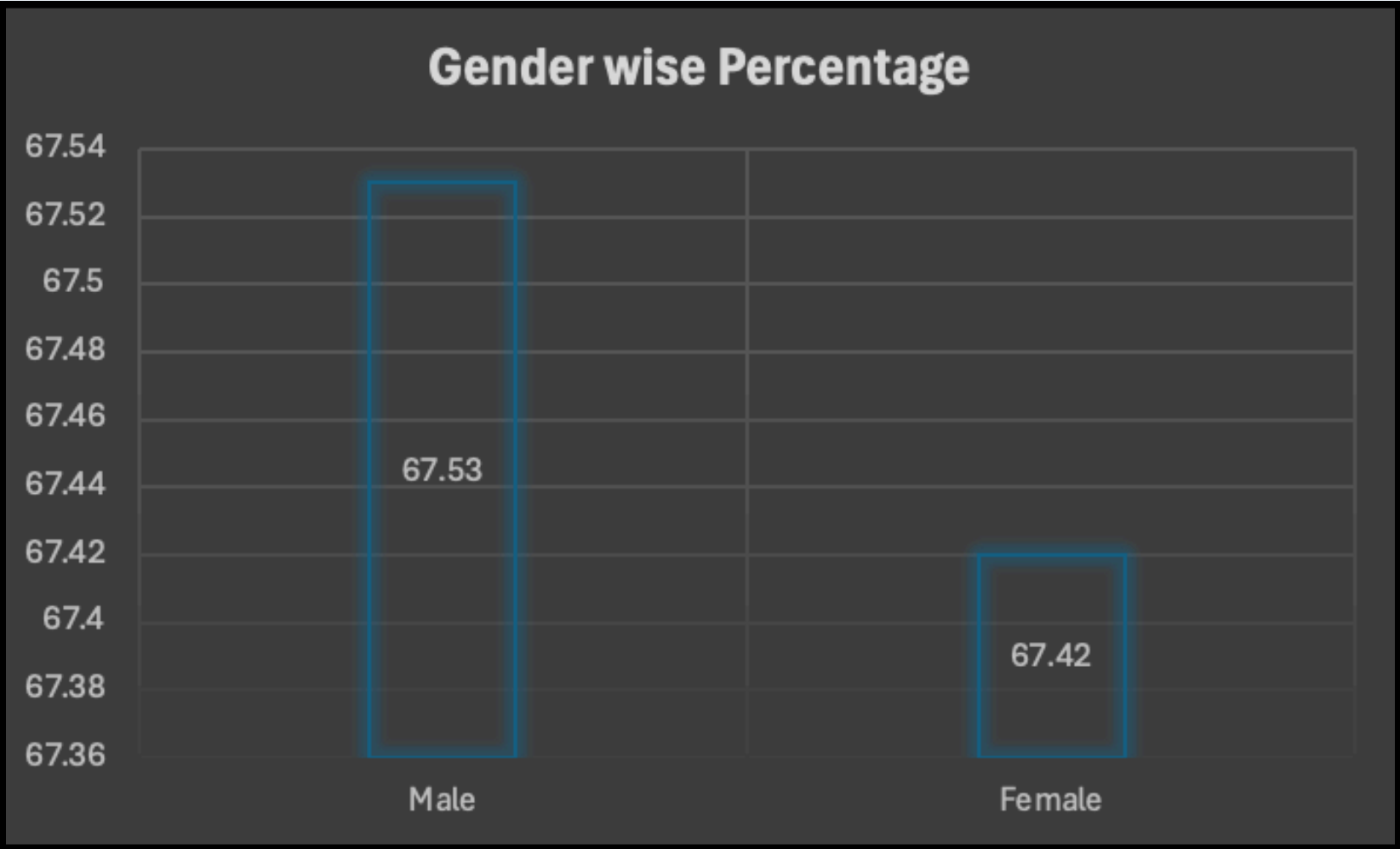
Performance Distribution by Student Demographics

How many students are Pass and Fail?



*The majority of students are in the **Pass** category.*

What is the average Final Percentage by Gender?



The average Final Percentage differs between male and female students by percentage points.

Class-wise Academic Performance

Which Class has the highest average Final Percentage?

Highest Performing Class

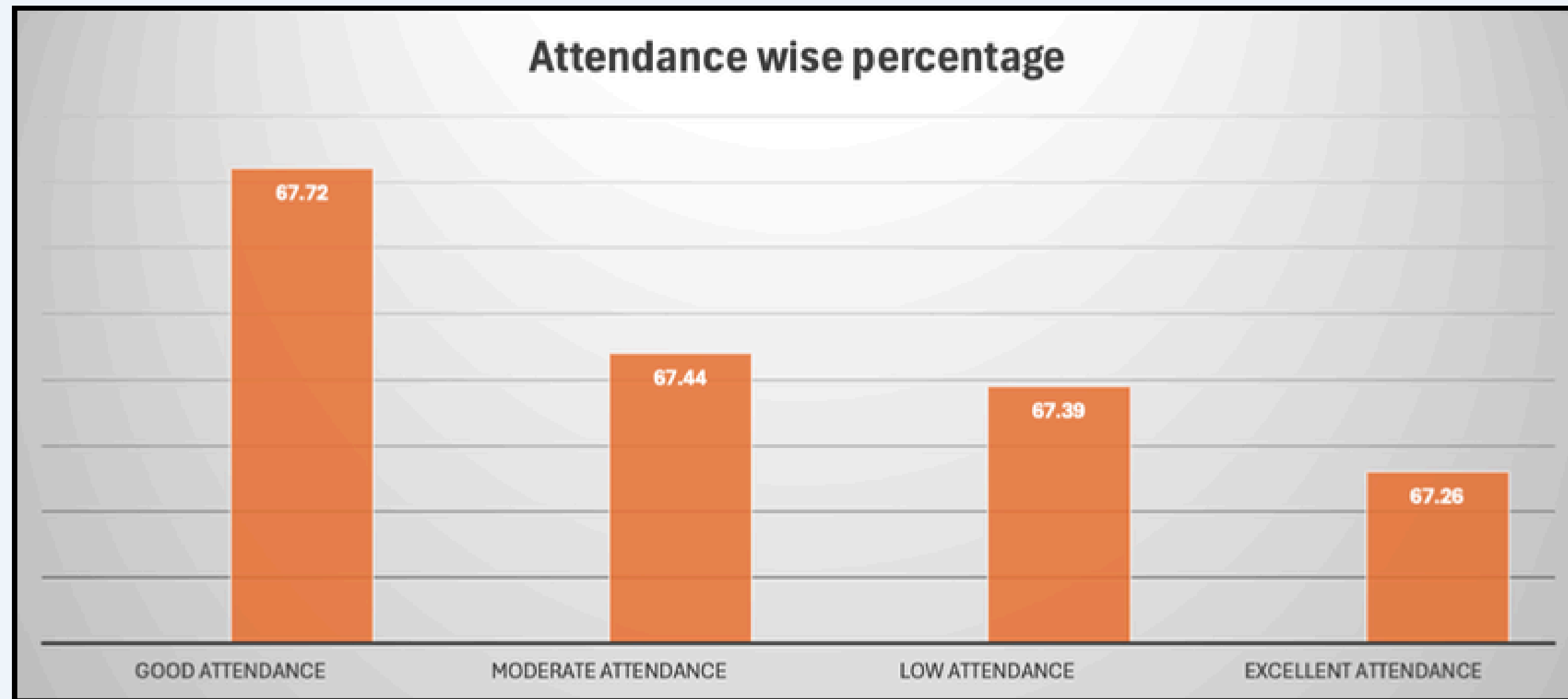
Class 9

67.75%

Class 9 recorded the highest average Final Percentage among the classes analyzed.

Attendance & Academic Performance

Does Attendance Relate to Academic Performance?

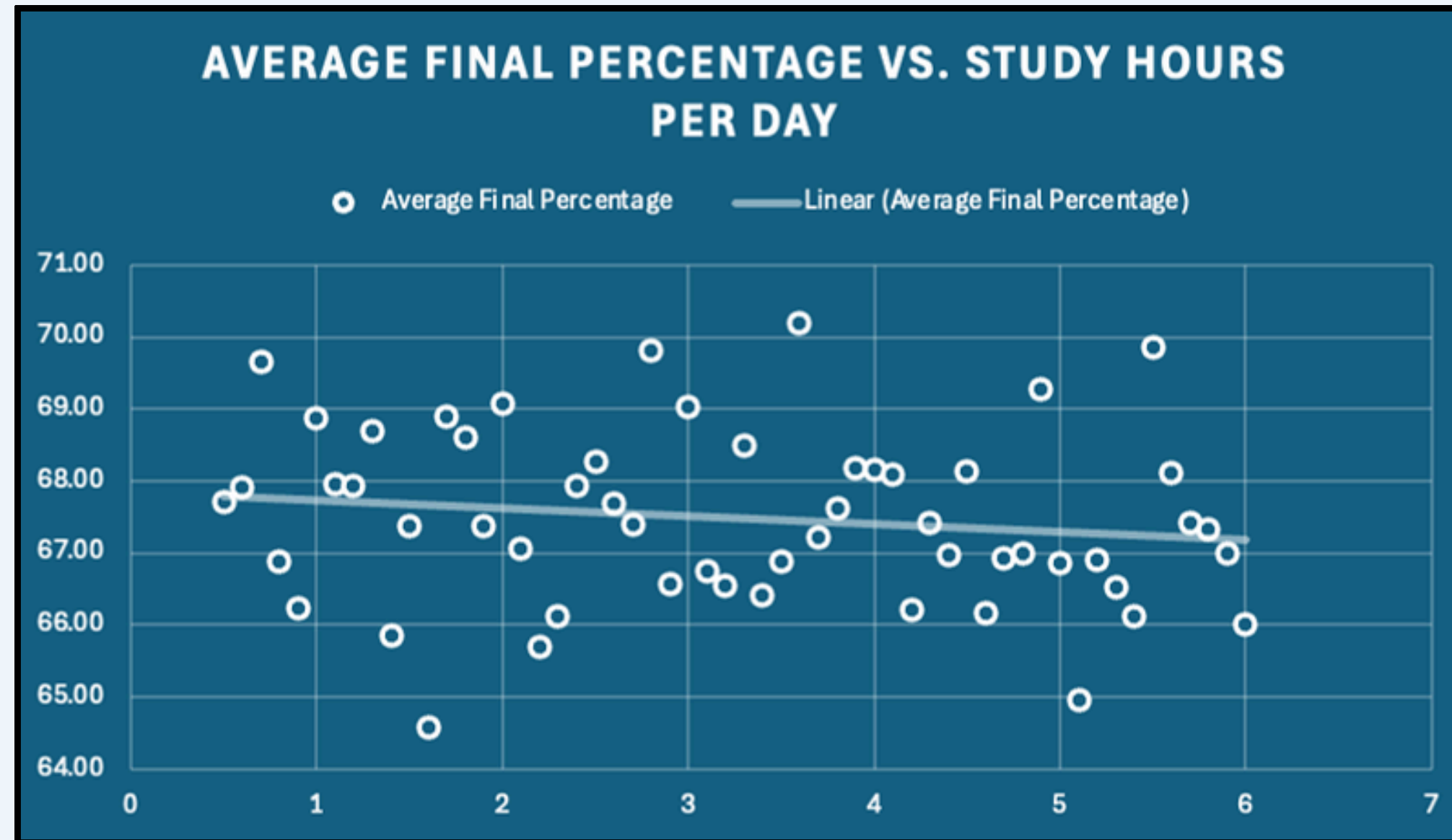


Students with higher attendance showed **67.72%** average Final Percentage compared with students having lower attendance.

This analysis identifies an association: it does not establish causation.

Study Hours & Final Percentage

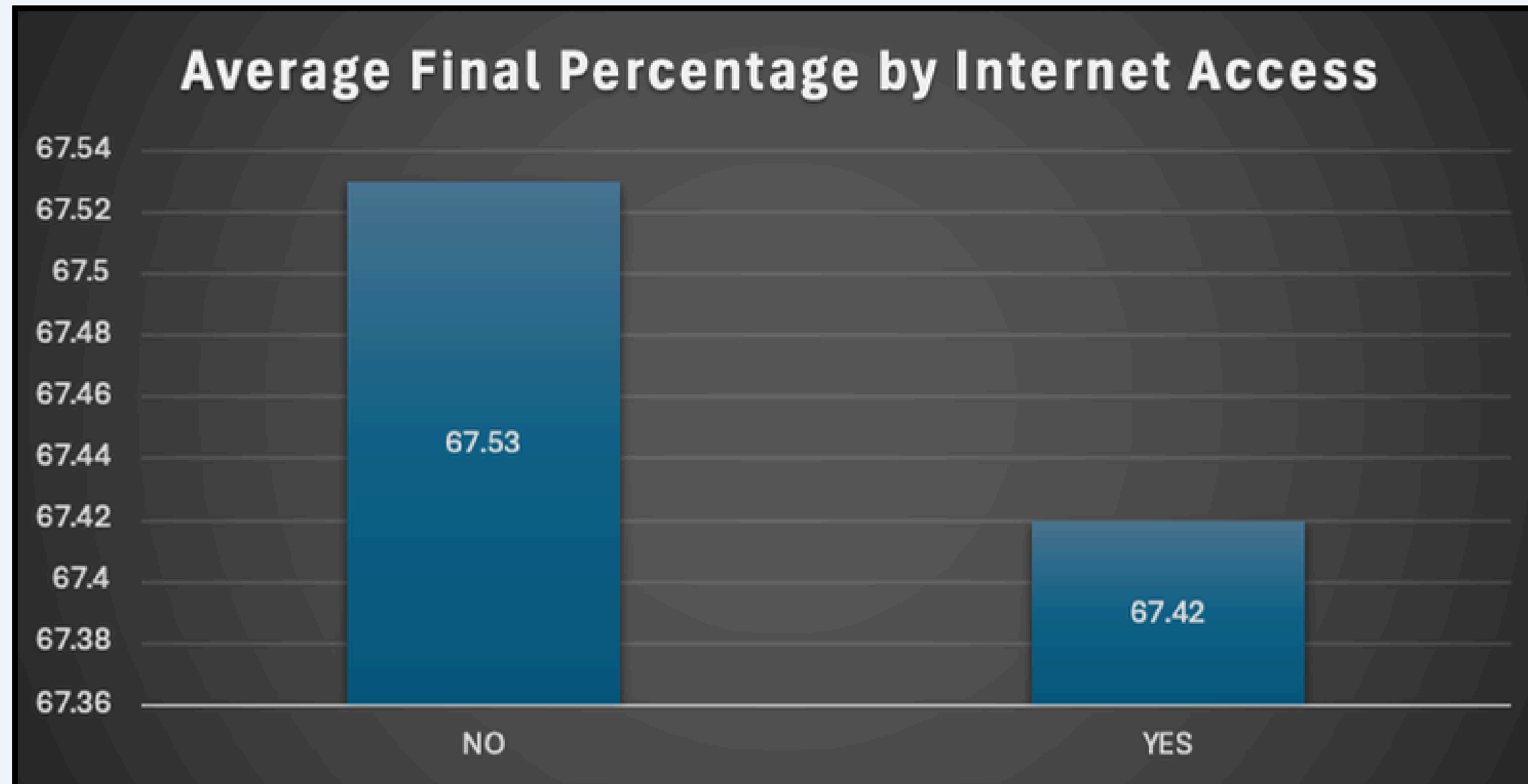
Study Hours vs. Final Percentage



More study hours did not consistently correspond to higher Final Percentage.

Internet Access & Performance

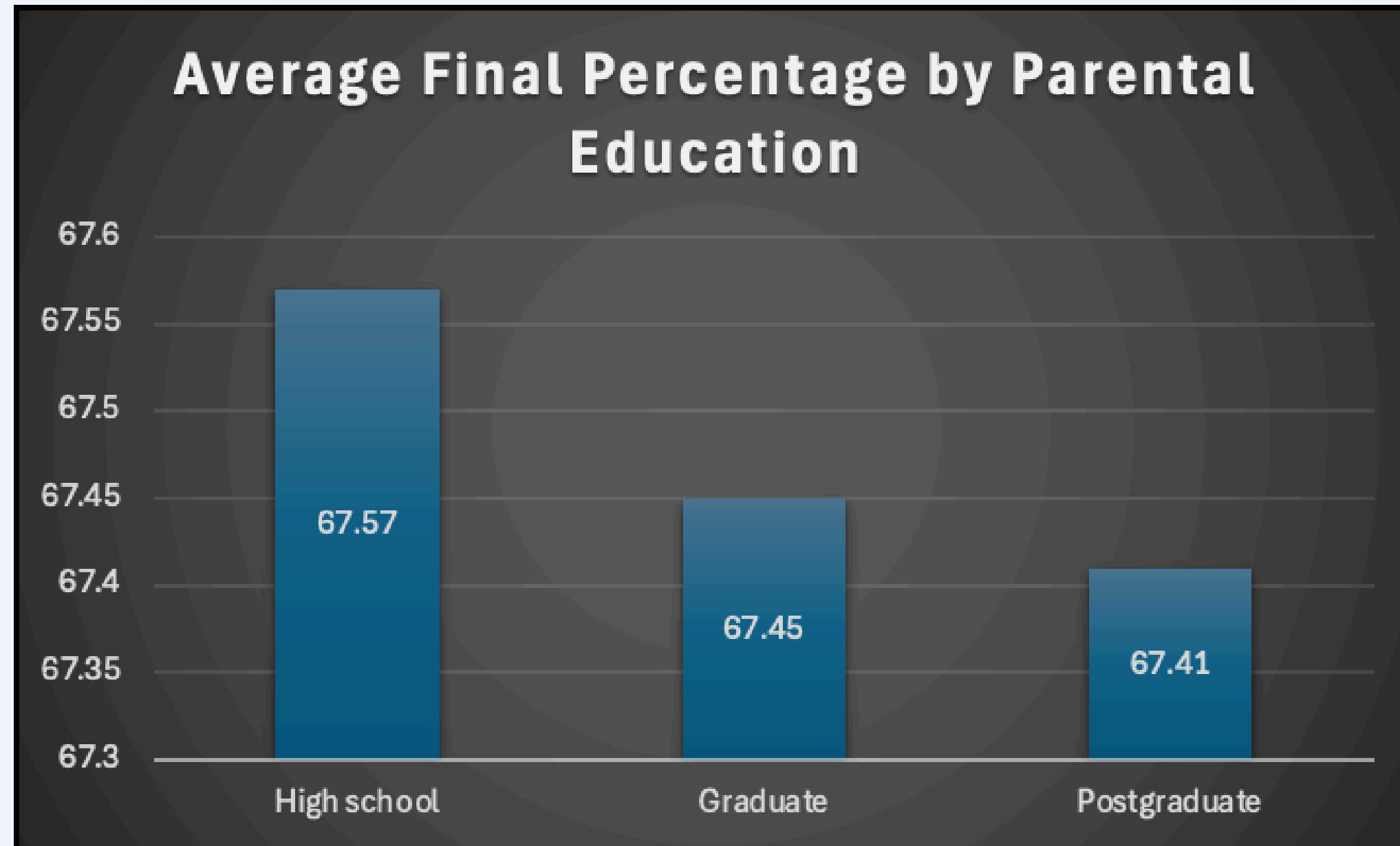
Impact of Internet Access on Academic Performance



*Students with Internet Access recorded a **67.53%** average Final Percentage compared with students without Internet Access.*

Parental Education & Performance

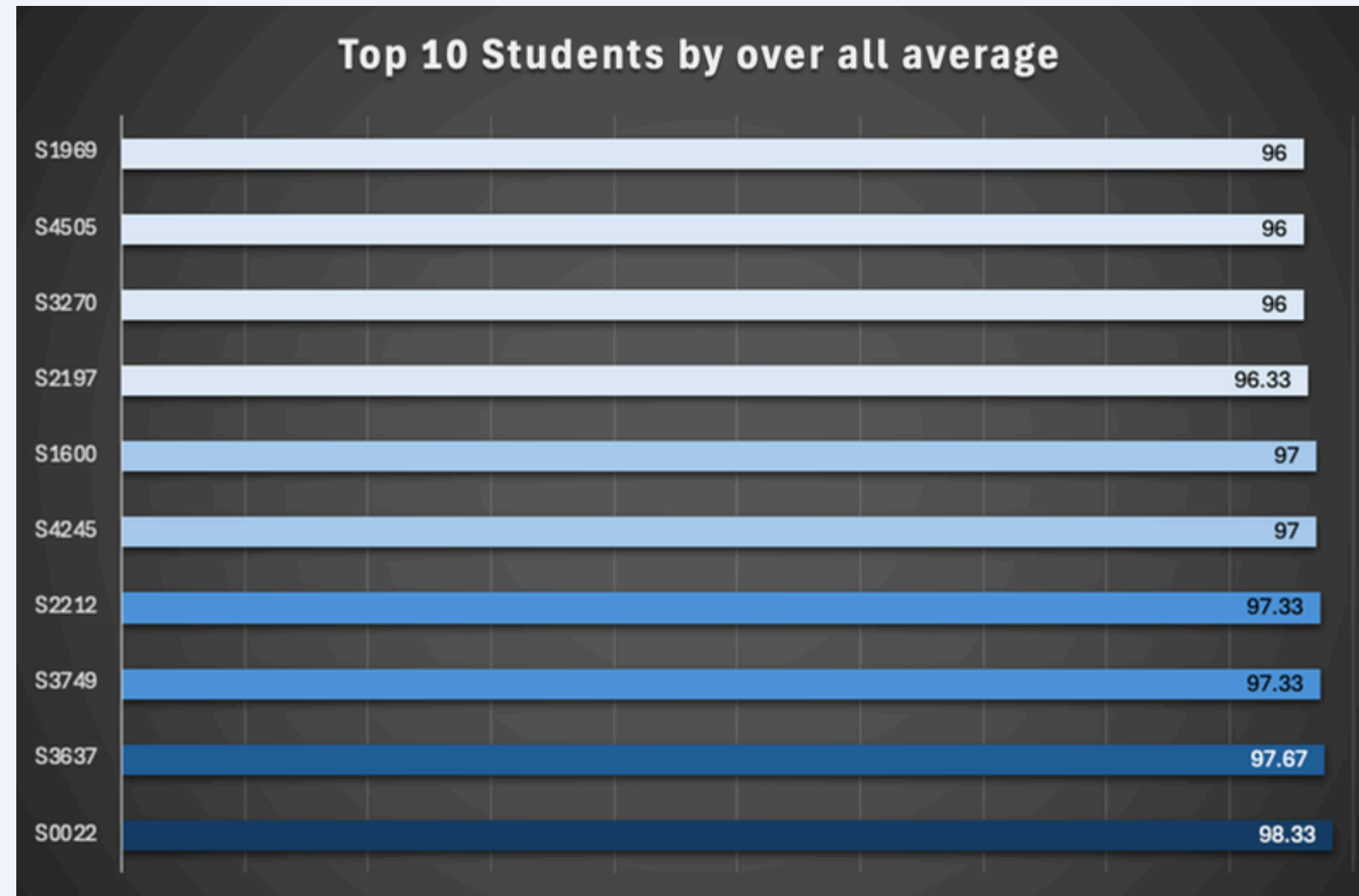
Parental Education and Student Performance



Average Final Percentage varies across different levels of parental education in the dataset.

Top 10 Students by Final Percentage

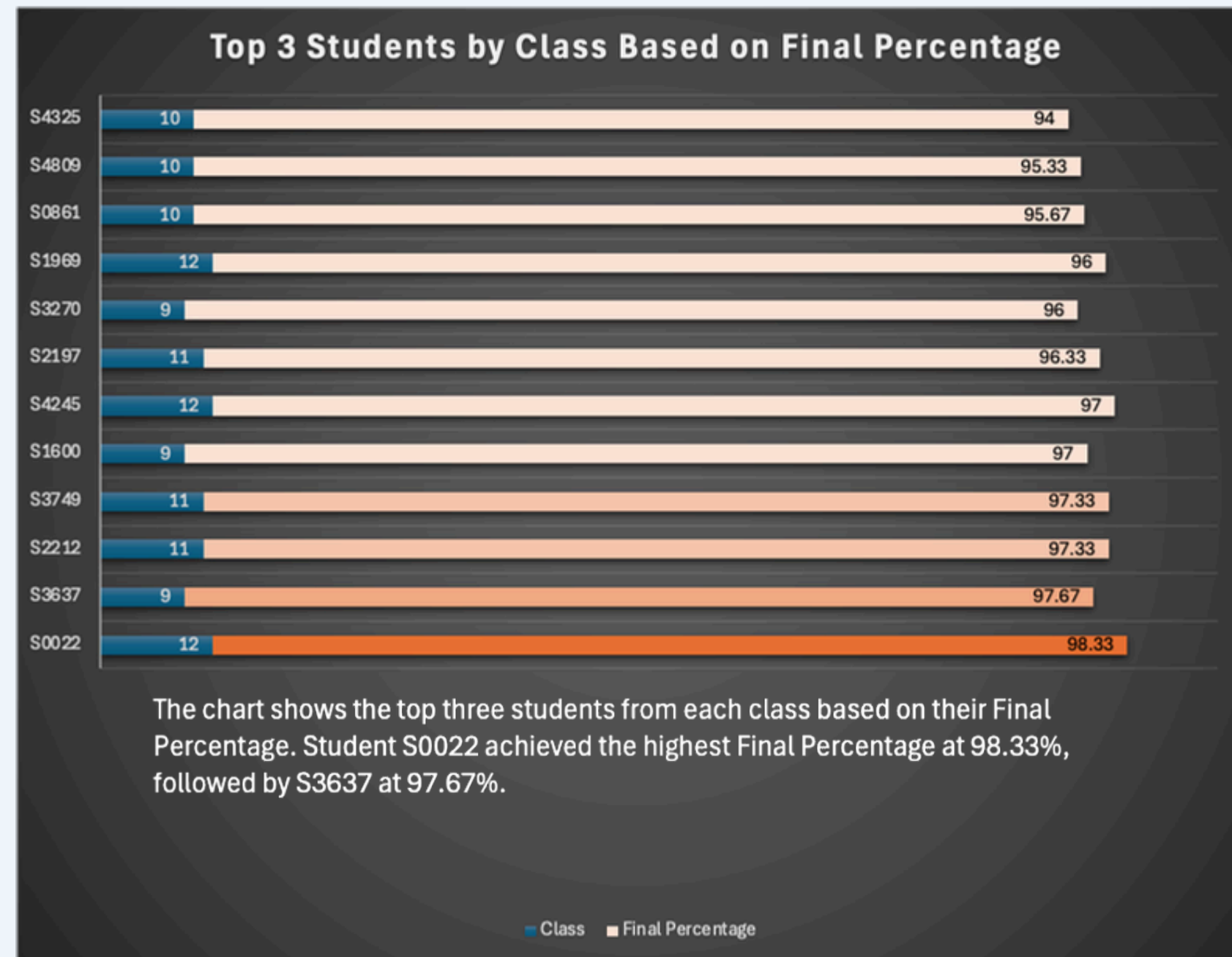
Students with the Highest Final Percentage



S0022 achieved the highest Final Percentage of 98.33% among the top 10 students.

Class-wise Student Ranking

Top 3 Students by Class



Highest Final Percentage:
S0022 — 98.33%

Students were ranked independently within each class using the SQL RANK() window function.

Key Findings

01. Overall Performance

Average Final Percentage
67.48%

02. Attendance

Higher attendance was associated with **Good** performance.

03. Study Hours

The relationship with Final Percentage was weak and slightly negative.

04. Internet Access

Students **with access** had **67.42%** average performance.

05. Parental Education

Performance varied across parental education groups.

06. High Performers

SOO22, students scored above the overall average.



SQL Techniques Applied

Show:

**AVG()
COUNT()
GROUP BY
CASE
Subqueries
Window Functions
RANK()
PARTITION BY
ORDER BY**

Eg:

01.

```
SELECT Student_ID, Final_Percentage
FROM student_performance_dataset
WHERE Final_Percentage > (
    SELECT AVG(Final_Percentage)
    FROM student_performance_dataset
)
ORDER BY Final_Percentage DESC
LIMIT 10;
```

02.

```
SELECT
    Student_ID,
    Final_Percentage
FROM student_performance_dataset
WHERE Final_Percentage > (
    SELECT AVG(Final_Percentage)
    FROM student_performance_dataset
);
```

03.

```
SELECT
    Parental_Education,
    AVG(Final_Percentage) AS
    Average_Final_Percentage
FROM student_performance_dataset
GROUP BY Parental_Education
ORDER BY Average_Final_Percentage
DESC;
```

04.

```
SELECT
    Internet_Access,
    AVG(Final_Percentage) AS
    Average_Final_Percentage
FROM
    student_performance_dataset
GROUP BY Internet_Access;
```

Recommendations

01

Monitor student attendance and academic performance together.

02

Identify students performing below the overall average.

03

Encourage effective study habits.

04

Use performance data to provide targeted academic support.

05

Consider additional factors in future analysis.

06



Limitations & Future Research

Limitations

- 01 Dataset contains limited variables.
- 02 Relationships do not necessarily imply causation.
- 03 Results may not represent all students.
- 04 Some external factors may be missing.

Future Research

- 01 Socioeconomic background
- 02 Family income
- 03 Teacher effectiveness
- 04 School resources
- 05 Subject-wise performance
- 06 Previous academic history
- 07 Correlation between multiple factors

Conclusion

This project demonstrates how SQL and Excel can be used together to transform student data into meaningful insights. The analysis examined overall performance, attendance, study habits, Internet Access, Parental Education and class-wise rankings. The findings provide a data-driven view of academic performance and demonstrate the usefulness of SQL-based analysis in educational research.



From Raw Data

SQL Analysis

Visualization

Research Insight

Thank You



Questions & Discussion

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